



BLACK HOLE HAIR

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Abstract

In this thesis, different black hole solutions of Einstein's Field Equations and the uniqueness of the Schwarzschild solution are examined. The concept of hair on a black hole is then defined, and several proofs of no-hair theorems for different fields on black hole backgrounds are reviewed. The proof of the No Short Hair theorem for black holes is also considered. Subsequently, various fields are placed outside a general spherically symmetric static metric, and the presence of hair is investigated; scalar fields, electromagnetic fields, Proca fields, and strongly coupled scalar fields are studied. Finally, a paper discussing possible observational evidence of black hole hair is reviewed.

Introduction

- The Einstein Field equations give us the solution for the metric to describe a given spacetime.

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi T_{\mu\nu}$$

- The three known solutions are Schwarzschild, Kerr, Reissner-Nordstrom metric. The parameters for each of them is mass, charge and Angular momentum.
- Birkoff, Isreal, Wald and Carter have given uniqueness theorems for these solutions. This implies we cannot find any other solutions, therefore any other parameters describing black holes. Note that these are all vacuum solutions.
- However, if we are able to find a metric solution which can support a field(scalar, vector or tensor) outside the horizon and it deviates from the known solutions, then we say that hair exists.
- J.D. Bekenstein showed that Scalar field ϕ with potential dependent on ϕ cannot exist outside

$$S_\phi = -\frac{1}{2} \int [\nabla_\alpha \phi \nabla^\alpha \phi + V(\phi^2)] \sqrt{-g} d^4x$$

This proof works as long as the metric is static and diagonal.

- Daniel Sudarsky showed that Higgs hair cannot exist in the Einstein theory. The condition here was that the Higgs field should have a semi definite positive potential.

$$S = \sum_i \int [\frac{1}{\alpha} R - \frac{1}{2} \nabla^\mu \phi_i \nabla_\mu \phi_i - V(\phi)] \sqrt{-g} d^4x$$

They also use the conditions that the spacetime is static and utilize the regularity of the horizon in the proof.

- Daniel Sudarsky also proposed that the presence of Non-linear matter in a black hole exterior can give rise to hair but it cannot be short, i.e. it has to be able to be detected atleast outside the distance of 1.5 times the event horizon. The proof works for matter fields that follow Weak Energy condition which basically says that the energy density observed should be non-negative.

Key Results

- Electromagnetic Field on a general static spherically symmetric metric: Radial magnetic cannot exist(attributed to the non-existence of magnetic monopoles). The metric becomes Reissner Nordstrom Solution.
- Proca field (massive EM field) outside general spherically symmetric metric. No Proca hair can exist
- Scalar hair on Reissner Nordstrom Black Hole. No scalar hair can exist.
- Non-minimally coupled scalar field with general static spherically symmetric metric background.

$$S = \int d^4x \sqrt{-g} [(\frac{-1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi) (\frac{-1}{2} m^2 \phi^2) + (\zeta R \phi^2)]$$

E.O.M:

$$\square \phi + (m^2 - 2\zeta R) \phi = 0$$

Multiplying with ϕ and integrating over spatial volume at a constant time slice will give:

$$\int_V -[g^{ij} \nabla_i \phi \nabla_j \phi + \phi^2 (m^2 - 2\zeta R)] \sqrt{-g} d^3x + \int_{\partial V} \phi \nabla^\alpha \phi d\Sigma_\alpha = 0$$

Boundary conditions reduce the integral to slice will give:

$$\int_V -[g^{ij} \nabla_i \phi \nabla_j \phi + \phi^2 (m^2 - 2\zeta R)] \sqrt{-g} d^3x = 0$$

- When $R > 0$ and $R > \frac{m^2}{2\zeta}$ Then the integrand is not positive and therefore we can hope to find non trivial solutions for ϕ

$$R = \frac{8\pi((\partial_r \phi)^2 + 2m^2 \phi^2 - 6\zeta \square \phi^2)}{1 - 16\pi\zeta \phi^2}$$

- I have looked at the asymptotic solutions, i.e. Long hairs. I have assumed $\delta = \frac{1}{r^p}$ for asymptotic flatness and truncated the powers of R at $\frac{1}{r^4}$, $\frac{1}{r^3}$, $\frac{1}{r^2}$ and $\frac{1}{r}$ to get differential equations of ϕ .

Method

- We work with a general static spherically symmetric metric

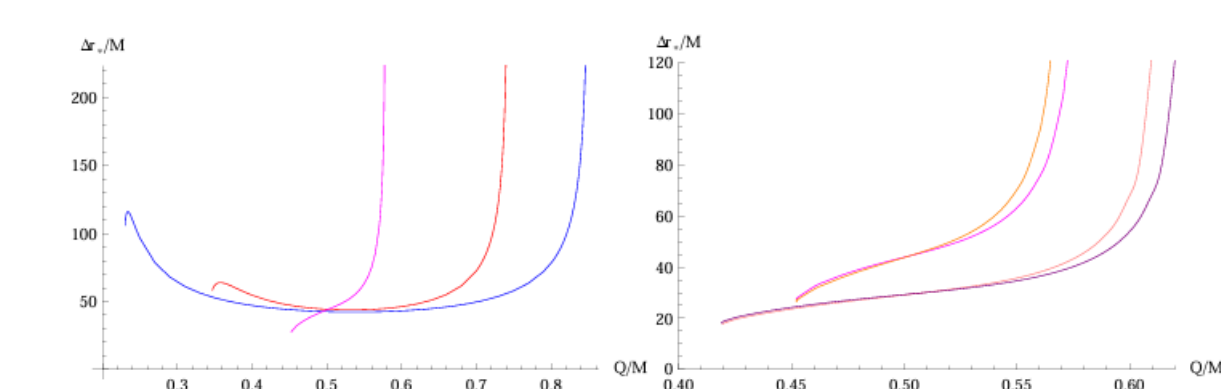
$$ds^2 = -e^{-2\delta(r)} \mu(r) dt^2 + \frac{1}{\mu(r)} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

where $\mu = 1 - \frac{2m}{r}$

- We place a field of our choice on the exterior of the horizon and check that a non trivial time independent profile of this field can be supported while giving us a non constant δ function or a deviation in the functional form of $\mu(r)$ from the regular Schwarzschild solution.

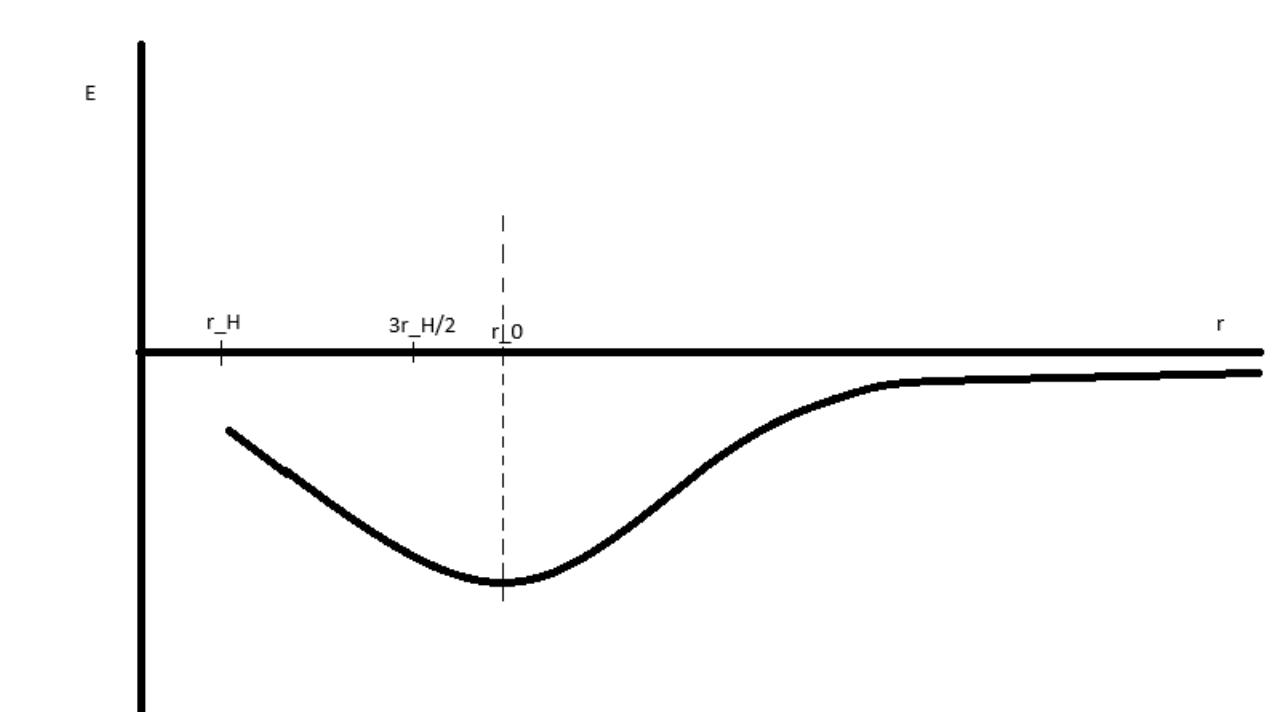
Observational Evidence

- Primary hairs are those that are completely independent of Mass, Charge and Angular Momentum.
- Einstein-Gauss Bonnet Theory can include scalar and vector in the action.
- The metric function $f(r)$ is dependent on the mass of the black hole M and the hair parameter Q .
- There are two peaks in the potential of the scalar perturbations, which are caused by the non-monotonic behaviour of $f(r)$. These peaks cause echoes in Gravitational waves caused during a Black Hole merger.



No Short Hair

We define a function $\epsilon = e^{-\delta} r^4 T_r^r$. We argue that this function varies as following:



From here we conclude that the field must be atleast detected outside $\frac{3r_H}{2}$

Future Works

- Find non-trivial solutions for the above problem
- Analyse if hair can look enhanced for different observers.

References

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